ALESSIO CURCIO

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Artificial Intelligence & Robotics Engineer | M.Sc. Student @ Sapienza University of Rome

PROFESSIONAL SUMMARY

Software Engineer and M.Sc. student in **Artificial Intelligence & Robotics (AIRO)** at **Sapienza University of Rome**, holding a Bachelor's degree in **Applied Computer Science and Artificial Intelligence (ACSAI)** taught entirely in English. Maximum marks (30 and 30 cum laude) in university programming examinations (Python & Java). Technical expertise focuses on **Multi-Agent Orchestration**, **Privacy-First Local LLMs**, **Real-Time Computer Vision (YOLO)**, and **Concurrent Robotic Systems**. Strong competitive problem-solving track record across national Mathematics and Informatics Olympiads and top rank in international algorithmic competitions.

EDUCATION

Master of Science (M.Sc.) in Artificial Intelligence and Robotics — **2024 – Present**
Sapienza University of Rome — Faculty of Information Engineering, Informatics and Statistics — *Rome, Italy*

- **Curriculum:** Autonomous Robotics, Computer Vision, Multi-Agent Systems, Reinforcement Learning, Robot Programming.

Bachelor of Science (B.Sc.) in Applied Computer Science and AI (ACSAI) — **Graduated**
Sapienza University of Rome — Degree Program Taught 100% in English — *Rome, Italy*

- **Academic Honors:** Scored 30 and 30 cum laude in University Programming Examinations (Python and Java).
- **Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Web & Software Architecture, Computer Networks, Artificial Intelligence, Concurrent & Multicore Programming.

Scientific High School Diploma — Applied Sciences & Mathematics Curriculum — **Grade: 98 / 100**
Liceo Scientifico Statale 'Vito Volterra' — *Graduated 2023*

TECHNICAL SKILLS

Artificial Intelligence & Vision:	PyTorch, TensorFlow, Keras, YOLOv11, OpenCV, Local LLMs (LM Studio), Hugging Face, Qwen3-TTS, Scikit-Learn, Reinforcement Learning.
Programming Languages:	Python (Primary / Advanced), Java, C, C++, Go (Golang), TypeScript, JavaScript, SQL, Bash, R.
Backend & Distributed Systems:	Go, Docker, Docker Compose, Node.js, WebSockets, PostgreSQL, REST APIs, Multithreading (POSIX).
Frontend & UI Development:	React, Vue.js, TailwindCSS, HTML5/CSS3, PyQt5, Tkinter, Flet (Flutter).
Tools & Platforms:	Git, GitHub, Linux Environments, Tesseract OCR, Streamlit, IBM Cloud & Watson.

KEY TECHNICAL PROJECTS (GITHUB: CRUCIO104)

OpenSyntax AI Coding Workspace — *TypeScript, Multi-Agent Orchestration, React, Local LLMs, Node.js*

- Architected a next-generation AI workspace featuring **multi-agent orchestration** for autonomous software development.
- Supported dynamic multi-file project scaffolding and real-time compilation across Python, C/C++, Java, Go, and Rust.
- Engineered hybrid execution support for both privacy-preserving local LLMs (via LM Studio) and cloud endpoints (OpenAI).

Privacy-First Local LLM Email Agent — *Python, LM Studio, Desktop GUI, Local AI Automation*

- Developed an autonomous desktop inbox assistant operating 100% locally to eliminate sensitive data transmission to external clouds.
- Integrated LM Studio inference to produce structured email summaries, automated action items, and rule-based priority tagging.

Echo Real-Time Messaging Platform — *Go (Golang), Vue.js, WebSockets, Docker Compose, PostgreSQL*

- Designed and deployed a high-performance messaging architecture for Sapienza's Web & Software Architecture examination.
- Implemented persistent WebSocket duplex communication for 1-on-1 chats and multi-user channels with full Docker containerization.

League of Legends Autonomous Bot (YOLOv11) *Python, YOLOv11, Computer Vision, PyTorch, Screen Analysis*

- Engineered a real-time computer vision agent utilizing **YOLOv11** for minion detection and priority classification in live gameplay.
- Executed automated target prioritization, skillshot mechanics, automated level-ups, and turret proximity safety routines.

Voice Cloning & Speech Synthesis Interface (Qwen3-TTS) *TypeScript, React, Docker, Speech AI, Qwen3-TTS*

- Built a high-fidelity web client for zero-shot voice cloning and multilingual speech synthesis using Qwen3-TTS.
- Delivered sub-second audio generation with responsive modern UI and modular Docker container orchestration.

Financial Stock Market Predictor (LSTM) *Python, LSTM Neural Networks, Streamlit, Pandas, Yahoo Finance API*

- Implemented deep learning time-series forecasting models using Long Short-Term Memory (LSTM) recurrent neural networks.
- Provided real-time ticker ingestion, moving average technical indicators, and interactive predictive dashboard via Streamlit.

PDF File Extractor & OCR Desktop Suite *Python, PyQt5, Tesseract OCR Engine, DOCX/TXT Export*

- Created an intuitive cross-platform desktop application extracting digital and scanned image PDF text with high OCR accuracy.
- Integrated multi-language OCR recognition and structured document conversion with a modern dark theme interface.

Concurrent & Robotic Systems Programming *C, C++, POSIX Threads, Mutex Synchronization, Robotic Kinematics*

- Implemented low-level multi-threaded concurrent programs, thread-safe memory allocations, and synchronization primitives.
- Modeled robotic kinematic transformations and sensor feedback processing algorithms for autonomous actuators.

INDUSTRIAL COLLABORATIONS & RESEARCH

Technical Research & Big Data Analytics — IBM & Sapienza University ***Academic Technical Activity***

- Conducted research in Big Data analytics for psychological and behavioral profiling models utilizing public data streams.
- Designed and implemented customer support conversational chatbots leveraging IBM Cloud services and IBM Watson AI.

Software Engineering & Digital Acceleration Labs — Ericsson & Fondazione Mondo Digitale ***Technological Program***

- Participated in hands-on engineering challenges in telecommunications architectures, advanced object-oriented design, and robotics.

COMPETITIONS, HONORS & CERTIFICATIONS

- **TeraBebras International Competition:** Attained the **maximum possible score** in algorithmic problem-solving and computational design.
- **Mathematics & Computer Science Olympiads:** Qualified for regional and national stages in Italian Mathematics, Informatics and Problem Solving Olympiads.
- **Bocconi University Mathematics Games:** Advanced through competitive stages to national ranking rounds.
- **National Cybersecurity Competition:** Successfully cleared qualification stages in algorithmic security and cryptography challenges.
- **Languages & International Experience: Italian** (Native), **English** (CEFR B2 Certification, Academic English B.Sc. degree, Oxford study abroad program).